"Semester project"

Title: "The Maze Game"

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1. Introduction

This project is a 2D **Maze Game** developed using **Python** and **OpenGL** libraries. The player navigates a maze, collects coins, avoids enemies, and aims to reach the goal. This game demonstrates the use of computer graphics for interactive applications, emphasizing 2D rendering, animation, and game logic.

2. Objectives

✧ Develop a fully functional 2D maze game.

- ✓ **Collectibles (Coins)**
- ✓ **Enemies with Random Colors**
- ✓ **Scoring System**
- ✓ **Lives Counter**
- ✓ **Game Timer**
- ✓ **Win & Lose Conditions**
- ✓ **Restart Option**

3. Tools & Technologies

Technology	Description
Python	Programming Language
PyOpenGL	OpenGL wrapper for Python
GLUT	Window and event handling
NumPy	For numerical operations (optional, not required here)

4. Features of the Game

Feature	Description
Maze Generation	Predefined static maze layout
Player Movement	Controlled via arrow keys
Collectibles	Coins represented as **small yellow circles**
Enemies	**Triangular enemies**, each with a **random color**

Feature	Description
Scoring	+10 points for every coin collected
Lives	Start with 3 lives; colliding with enemies reduces 1 life
Timer	Displays elapsed game time
Win Condition	Reach the **goal (green square)**
Restart Function	Press 'R' to restart the game
Exit	Press 'Esc' to exit

5. Working of the Game

1. **Initialization**:

   - ✓ The maze is displayed.
   - ✓ Player starts at the entrance.
   - ✓ Collectibles and enemies are randomly positioned.

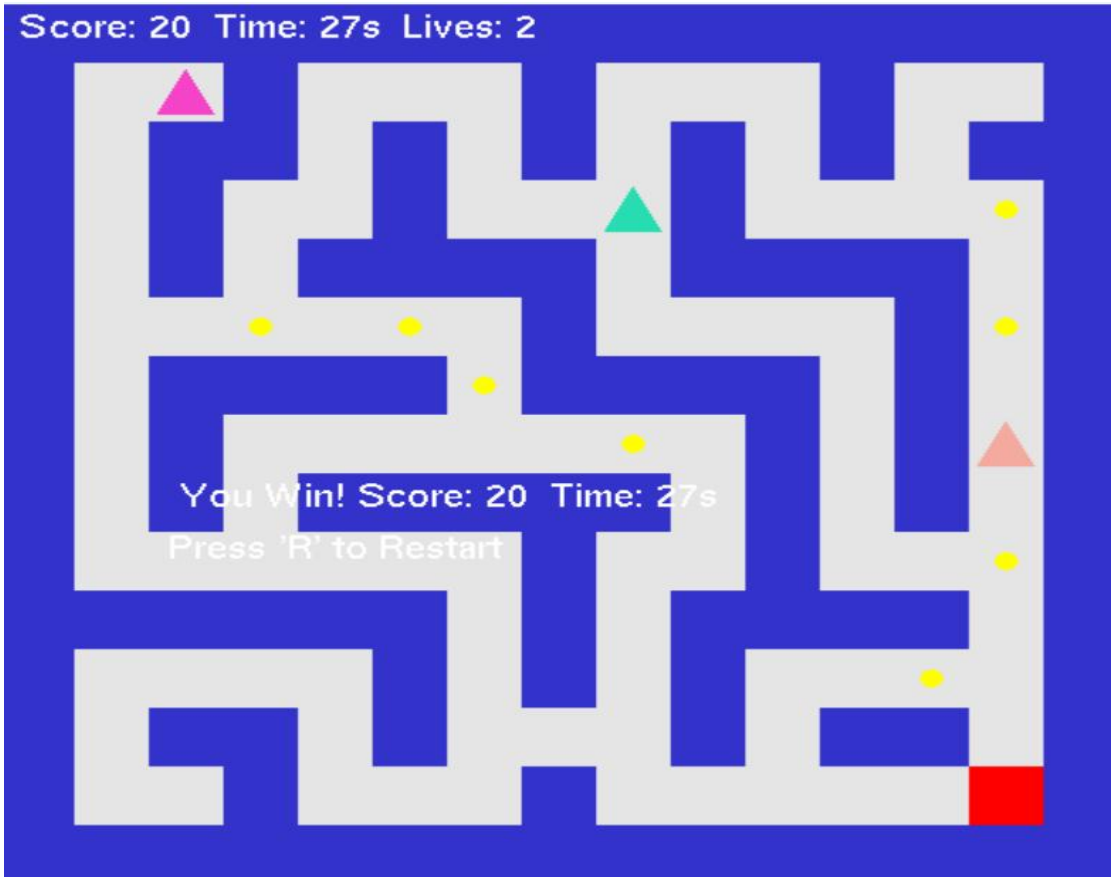
2. **Gameplay**:

   - ✓ The player uses **arrow keys** to navigate.
   - ✓ Coins increase the **score**.
   - ✓ Contact with enemies decreases **lives**.
   - ✓ Reaching the **goal** completes the game.
   - ✓ If **lives = 0**, "Game Over" is displayed.

3. **Restart/Exit**:

   - ✓ Press **'R'** to restart.
   - ✓ Press **'ESC'** to exit.

6. Screenshots



7. Conclusion

This project successfully demonstrates **2D graphics** programming in **Python with OpenGL**, applying fundamental computer graphics concepts such as:

- ✓ Coordinate transformations
- ✓ Color rendering
- ✓ Shape creation (Squares, Circles, Triangles)
- ✓ Event handling (keyboard input)
- ✓ Timer-based animation